

What Do Mathematicians Actually Do?

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TalkShop

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What is a Mathematician?

What do you think when you hear the term “mathematician?”
Maybe you think of ...

- Someone who likes math
- Someone who is good at math
- Someone who teaches math

A **mathematician** is dedicated to solving problems using a variety of knowledge based on extensive study of the field.

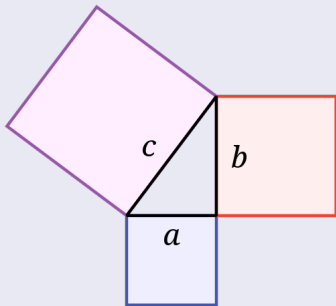
How Do Mathematicians Form Questions?

Just like other scientists, mathematicians pose problems they would like to answer. One way that problems are formed is by extending an existing question.

Pythagorean Theorem

The side lengths a , b , and c of a right triangle where c is the hypotenuse satisfies the following equation:

$$a^2 + b^2 = c^2.$$



How Do Mathematicians Form Questions?

Do we see any questions that we could pose appearing similar to the Pythagorean theorem?

Pythagorean Theorem

The side lengths a , b , and c of a right triangle where c is the hypotenuse satisfies the following equation:

$$a^2 + b^2 = c^2.$$

Maybe we ask...

- Are there positive integers a, b, c such that

$$a^3 + b^3 = c^3?$$

- Moreover, are there positive integers a, b, c such that

$$a^n + b^n = c^n$$

for some $n \geq 3$?

How Do Mathematicians Form Questions?

The quest to find positive integers a, b, c satisfying $a^n + b^n = c^n$ for $n \geq 3$ led to the following statement first posed by the mathematician Pierre de Fermat in 1637.

Fermat's Last Theorem

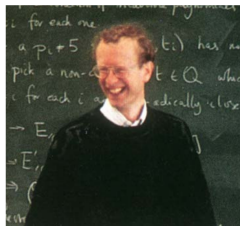
For $n \geq 3$, there are no positive integers a, b, c that satisfy

$$a^n + b^n = c^n.$$

While Fermat stated this as a **theorem**, it was not **proven** until 1995 by Andrew Wiles, more than 350 years later!



Pierre de Fermat



Andrew Wiles

What Do We Mean “Proof?”

In mathematics, statements are first posed as **conjectures**. It is only after mathematicians come up with a **proof** that covers every possibility we say a statement is true, and then it is called a **theorem**.

Just like other scientists, mathematicians can use data to help them form conjectures. Once we have a conjecture, the quest begins to find a proof.

We will see examples in the next few slides of examining data, forming a conjecture, and then proving the conjecture it is true.

 Data (1).png

Data $\longrightarrow$ Conjecture

Consider the values of $g(n) = n^2$.

n	n^2	n	n^2	n	n^2	n	n^2
1	1	6	36	11	121	16	256
2	4	7	49	12	144	17	289
3	9	8	64	13	169	18	324
4	16	9	81	14	196	19	361
5	25	10	100	15	225	20	400

Are there any patterns you notice?

- If n is even, then n^2 is even.
- If n is odd, then n^2 is odd.
- For each n , $(n + 1)^2 - n^2 = 2n + 1$.
For example, $6^2 - 5^2 = 36 - 25 = 11 = 2 \cdot 5 + 1$.

Conjecture $\longrightarrow$ Proof

If n is even, then n^2 is even.

Suppose n is even, so $n = 2k$ for some integer k . Then

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

Since $2k^2$ is an integer, this shows n^2 is even. $\square$

If n is odd, then n^2 is odd.

Suppose n is odd, so $n = 2k + 1$ for some integer k . Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k$ is an integer, this shows n^2 is odd. $\square$

Conjecture $\nrightarrow$ Proof

Not all conjectures turn out to be true: sometimes we find **counterexamples**. The mathematician Pierre de Fermat posed the following conjecture.

Fermat's Conjecture

The number $F(n) = 2^{2^n} + 1$ is prime. (Fermat primes)

This is true for the first few values:

$$\begin{aligned} F(0) &= 3, & F(1) &= 5, & F(2) &= 17, \\ F(3) &= 257, & F(4) &= 65,537 \end{aligned}$$

However, $F(5) = 4,294,967,297 = 641 \times 6,700,417$, so it is not prime. In fact, no other Fermat primes have ever been found. We can form a new conjecture:

Our Conjecture

For $n \geq 5$, the number $F(n) = 2^{2^n} + 1$ is **not** prime.

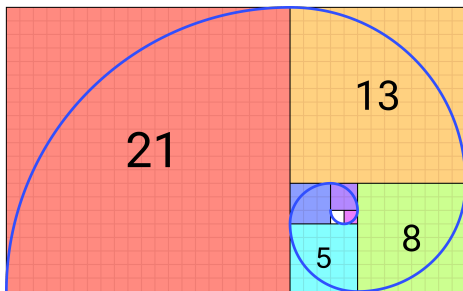
My Research Interests

I work in a branch of mathematics called **number theory**.

Number theorists study the integers and arithmetic functions with particular interest in prime numbers and mathematical “objects” that can be constructed from the integers.

The Fibonacci Sequence

Let $F_1 = 1$, $F_2 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 3$.



The Fibonacci Spiral

2-power Divisibility

A **rational number** has form a/b where a and b are integers. We will simplify a/b so that a and b have no common factors.

Definition

The **2-power divisibility** of a rational number a/b tells how many copies of 2 are in the numerator. In other words, if we write $a = 2^k \cdot a'$ where a' is not divisible by 2, then the 2-power divisibility of a/b is k . We will use the notation $\text{ord}_2(a/b)$ to denote the 2-power divisibility of a/b .

Examples

$$\textcircled{1} \quad \text{ord}_2\left(\frac{12}{5}\right) = \text{ord}_2\left(\frac{2^2 \cdot 3}{5}\right) = 2$$

$$\textcircled{2} \quad \text{ord}_2\left(\frac{32}{18}\right) = \text{ord}_2\left(\frac{2^5}{2 \cdot 3^2}\right) = \text{ord}_2\left(\frac{2^4}{3^2}\right) = 4$$

$$\textcircled{3} \quad \text{ord}_2(2048) = \text{ord}_2(2^{11}) = 11$$

2-power Divisibility

An exercise in number theory is to show the rational number

$$S(n) = \frac{2^1}{1} + \frac{2^2}{2} + \frac{2^3}{3} + \frac{2^4}{4} + \cdots + \frac{2^n}{n}$$

(it looks like a number when you find a common denominator) is highly divisible by powers of 2 as $n \rightarrow \infty$. Using our previous notation, this means that $\text{ord}_2(S(n))$ is getting larger as n does.

Examples

Let's look at the value of $S(8)$.

$$\begin{aligned} S(8) &= \frac{2^1}{1} + \frac{2^2}{2} + \frac{2^3}{3} + \frac{2^4}{4} + \frac{2^5}{5} + \frac{2^6}{6} + \frac{2^7}{7} + \frac{2^8}{8} \\ &= \frac{8192}{105} = \frac{2^{13}}{3 \cdot 5 \cdot 7} \end{aligned}$$

Thus $\text{ord}_2(S(8)) = 13$.

2-power Divisibility

Here are some tables showing values of $\text{ord}_2(S(n))$ for $n \leq 24$.

Recall that $S(n) = \frac{2^1}{1} + \frac{2^2}{2} + \frac{2^3}{3} + \frac{2^4}{4} + \cdots + \frac{2^n}{n}$.

n	$\text{ord}_2(S(n))$	n	$\text{ord}_2(S(n))$	n	$\text{ord}_2(S(n))$
1	1	9	9	17	17
2	2	10	10	18	18
3	2	11	10	19	18
4	5	12	12	20	21
5	8	13	12	21	22
6	5	14	12	22	21
7	5	15	12	23	21
8	13	16	22	24	27

Question: Are there any noticeable patterns?

Lower Bound on 2-power Divisibility

Theorem (S.)

For $n \geq 1$, $\text{ord}_2(S(n)) > \log_2(n)$.

In words, the 2-power divisibility of $S(n)$ will always be strictly larger than the base 2 logarithm of n . Here are tables that illustrate the theorem for $n \leq 16$.

n	$\text{ord}_2(S(n))$	$\log_2(n)$	n	$\text{ord}_2(S(n))$	$\log_2(n)$
1	1	0	9	9	3.17
2	2	1	10	10	3.32
3	2	1.58	11	10	3.46
4	5	2	12	12	3.58
5	8	2.32	13	12	3.70
6	5	2.58	14	12	3.81
7	5	2.81	15	12	3.91
8	13	3	16	22	4

My thesis work focuses on finding a better (much closer to the actual value) lower bound on $\text{ord}_2(S(n))$. Additionally, we look at functions similar to $S(n)$ to form new conjectures.

Conjecture

Let $T(n) = \frac{3^1}{1} + \frac{3^2}{2} + \frac{3^3}{3} + \cdots + \frac{3^n}{n}$. Then the 3-power divisibility of $T(n)$ (denoted $\text{ord}_3(T(n))$) gets large as $n \rightarrow \infty$.

This turns out to be **false**! If we check some data:

- $T(1) = 3 \implies \text{ord}_3(T(1)) = 1$
- $T(2) = \frac{15}{2} \implies \text{ord}_3(T(2)) = 1$
- $T(3) = \frac{33}{2} \implies \text{ord}_3(T(3)) = 1$
- $T(9) = \frac{987393}{280} = \frac{3 \cdot 11 \cdot 29921}{2^3 \cdot 5 \cdot 7} \implies \text{ord}_3(T(9)) = 1$

Thank you!

Not all
math jokes
are bad.
Just sum.



Theorem

The sum of two integers x and y with the same parity is even.

Proof. We have two possible cases.

Case 1: x and y are both even.

In this case, we can say $x = 2k$ and $y = 2\ell$ for some integers k and ℓ . Then

$$x + y = 2k + 2\ell = 2(k + \ell).$$

Since $k + \ell$ is an integer, this shows that $x + y$ is even.

Case 2: x and y are both odd.

In this case, we can say $x = 2k + 1$ and $y = 2\ell + 1$ for some integers k and ℓ . Then

$$x + y = 2k + 1 + 2\ell + 1 = 2k + 2\ell + 2 = 2(k + \ell + 1).$$

Since $k + \ell + 1$ is an integer, this shows that $x + y$ is even. □